

Unit 11, Lesson 1: Graphing Proportional Relationships

Benchmark

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

Learning Targets

- ☐ I can graph a proportional relationship from a table, an equation, or a written description.
- ☐ I can describe common features of the graphs of proportional relationships.

11.1.1 Warm-Up: Dancing with Math

Karin Ditchfield lived in three different countries growing up – Brazil, England, and Argentina – where three different languages are spoken. Learning new languages can be challenging, but she explains despite these challenges the language of math was always the same.

1. Karin says, “Math is not just about working with numbers. I see it as a global language that has a positive impact on the world.”

What are some reasons why math could be considered a global language?

2. Karin mentioned that growing up in Brazil meant growing up dancing. She says, “When you’re learning a new dance, practice makes perfect. It’s exactly the same with math.”

Explain whether you agree with this explanation.

11.1.2 Exploration: Graphs of Proportional Relationships

In this course, **proportional relationships** have been used to make predictions about populations from sample data, create scale drawings, convert between units of measure, and solve problems involving percentages.



A **proportional relationship** is a collection of pairs of numbers that are in equivalent ratios.

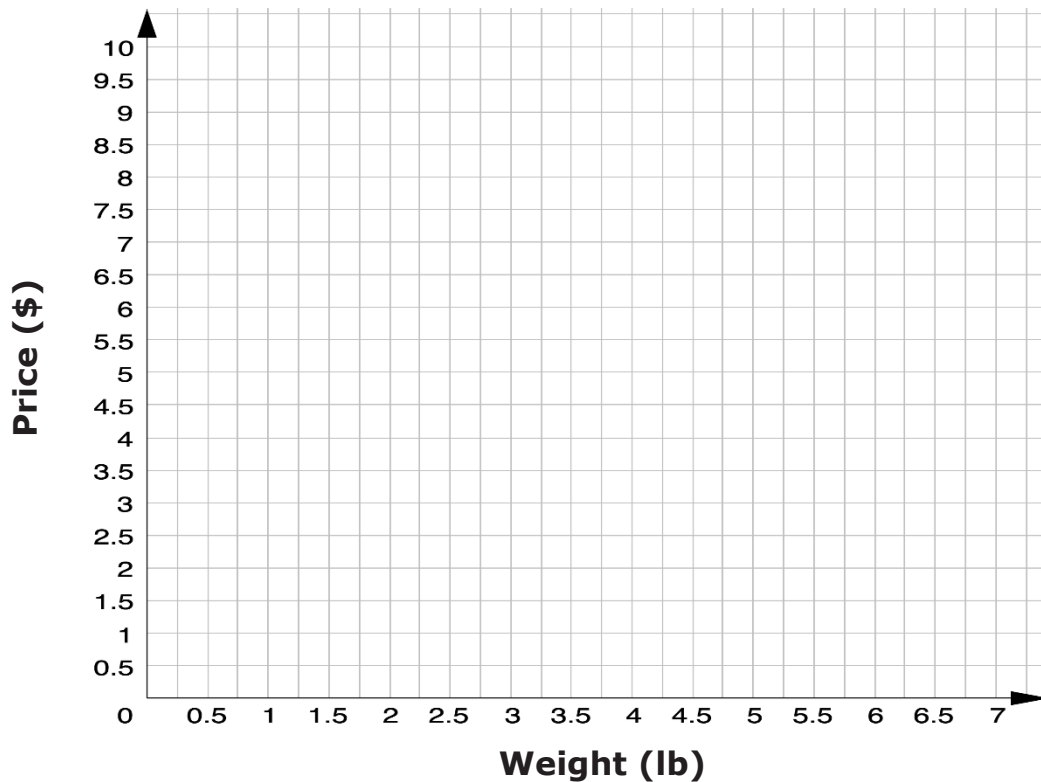
Proportional relationships can be represented by graphs, tables of values, equations, and written descriptions.

Work with your partner to complete the following.

1. Madelyn and her family went to a U-Pick farm in Lithia, FL to pick blueberries. The table shows the weight, in pounds (lb), and price, in dollars, of the bag of blueberries each family member picked.

Weight (lb)	Price (\$)
0.75	2.85
2.6	9.88
2	7.60
1.2	4.56

- a. What is the unit price per pound of blueberries?
- b. Explain how you determined the unit price.
- c. Graph the relationship by plotting a point to represent each family member's bag of blueberries.



- d. Plot a point on the graph to represent the unit price per pound of blueberries.
- e. Plot a point on the graph to represent the price for 0 pounds of blueberries.
- f. Connect the points plotted on the graph. What do you notice?
- g. Discuss with your partner whether the graph of the relationship between the number of pounds of blueberries picked and price extends beyond the points plotted on the graph in either direction.

h. Complete the statements.

The graph

☐ does

☐ does not

extend below the point (0,0) because it is not

possible to pick a

☐ positive

☐ negative

number of pounds of blueberries.

The graph

☐ does

☐ does not

extend above the point (2.6,9.88)

because it is possible to pick any

☐ positive

☐ negative

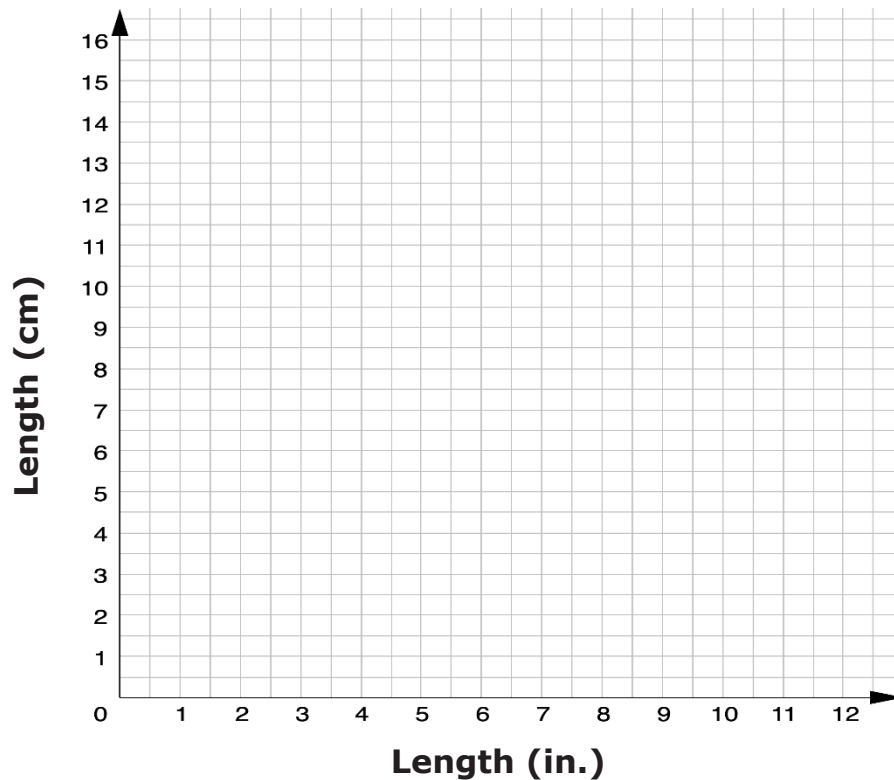
number of pounds of blueberries.

2. There are approximately 2.54 centimeters (cm) in every 1 inch (in.). This proportional relationship can be represented by the equation $c = 2.54n$, where c represents the length of an object in centimeters, and n represents the length of the same object in inches.

a. Use the equation to complete the table of values.

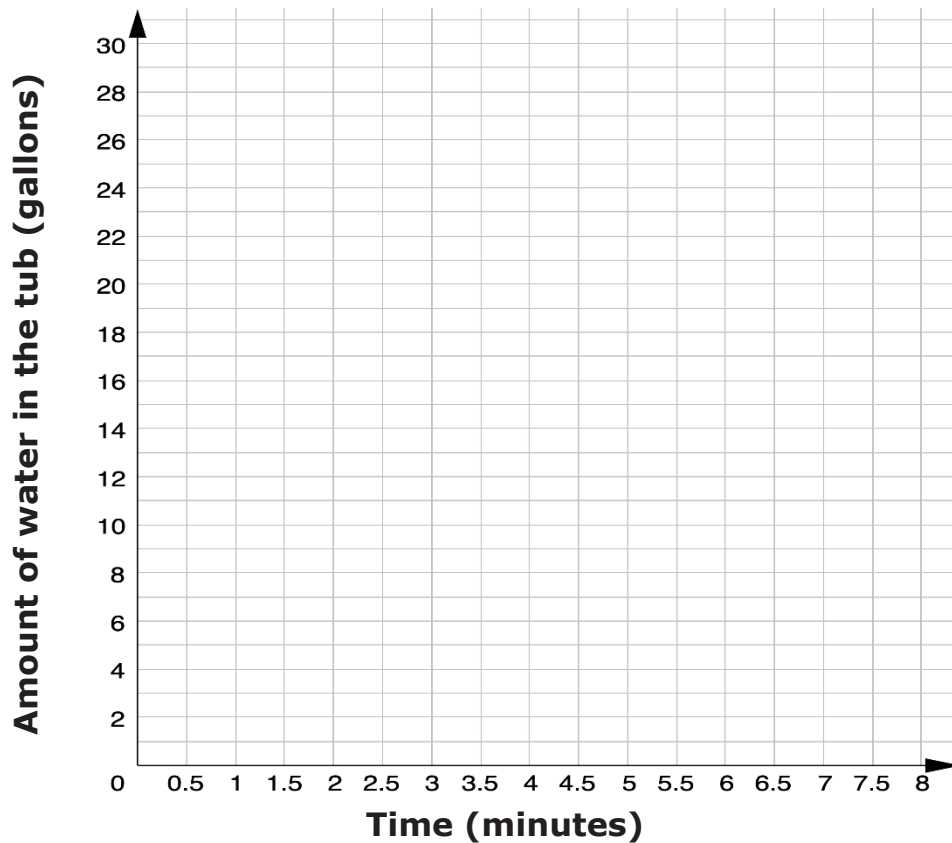
n length (in.)	c length (cm)
1	
6	
$3\frac{1}{2}$	
2.25	

b. Graph the ordered pairs from the table of values.



- c. Connect the points plotted on the graph to create a ray that represents all possible equivalent lengths.
 - d. Discuss with your partner whether there is a point on the line that represents the approximate number of inches that are equivalent to 5 centimeters. If there is such a point, draw an X to mark this point on the line. If there is not such a point, explain why not.
3. An empty bathtub is filling at a rate of 4.5 gallons per minute.
- a. How many gallons of water are in the tub after 0 minutes?
 - b. How many gallons of water are in the tub after 3 minutes?

- c. How many minutes does it take for the tub to fill to 31.5 gallons of water?
- d. Using the values determined in Parts A-C, represent the relationship of the amount of water in the tub, in gallons, with time, in minutes, on the coordinate grid.



- e. Connect the points plotted on the graph to create a ray.
- f. Complete the statement.

The bathtub is filling up at a constant unit rate of _____

per ☐ gallons
☐ minutes

☐ gallons
☐ minutes

- g. Place an X on the graph at the point that represents the unit rate described in Part F.

11.1.3 Bring It Together: Graphs of Proportional Relationships

Each graph created in the Exploration represents a proportional relationship. Graphs representing proportional relationships have certain features in common.

1. Complete each statement to describe the common features of the graphs of proportional relationships.

- a. Graphs of proportional relationships

☐ include
☐ do not include

the origin, $(0,0)$, because when the value

of one quantity equals 0, the value of the

other quantity ☐ will also
☐ will not

equal 0.

- b. Graphs of proportional relationships form

a ☐ line.
☐ curve.

- c. A unit rate for the proportional relationship can be

identified on the graph using the ☐ x -value
☐ y -value

of the point

where the ☐ x -value
☐ y -value

is 1.

11.1.4 Wrap-Up: Ticket Out the Door

1. The Environmental Protection Agency (EPA) estimated that in 2013, the average amount of garbage produced in the U.S. was 4.4 pounds (lb) per person per day.

The table of values represents the proportional relationship between the amount of garbage produced in the U.S., in pounds, per person, with time, in days.

Number of Days	Garbage per Person (lb)
1	4.4
3	13.2
4.5	19.8
5	22

Graph this proportional relationship.

